WASP-77Ab

R-1486

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-77_151118` · created 2026-08-01T13:25:48+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2457344.72342 +/- 0.00093 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.131 +/- 0.0046
Transit depth (Rp/Rs)^2	1.72 +/- 0.12 [%]
Orbital Inclination (inc)	87.98 +/- 1.21
Airmass coefficient 1 (a1)	1.1808 +/- 0.0059
Airmass coefficient 2 (a2)	-0.1197 +/- 0.0049
Scatter in the residuals of the lightcurve fit is	0.5 %
Best Comparison Star	None
Optimal Aperture	5.93
Optimal Annulus	12.34
Transit Duration (day)	0.0893 +/- 0.0018

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.131 ± 0.0046 vs 0.1301 (deviation 0.2σ)
Duration fit vs published	0.0893 d vs 0.0908 d (deviation 0.85σ)
Mid-time O–C	2.0 min
Depth SNR	28.5
Residual scatter	0.5 %

Light curve

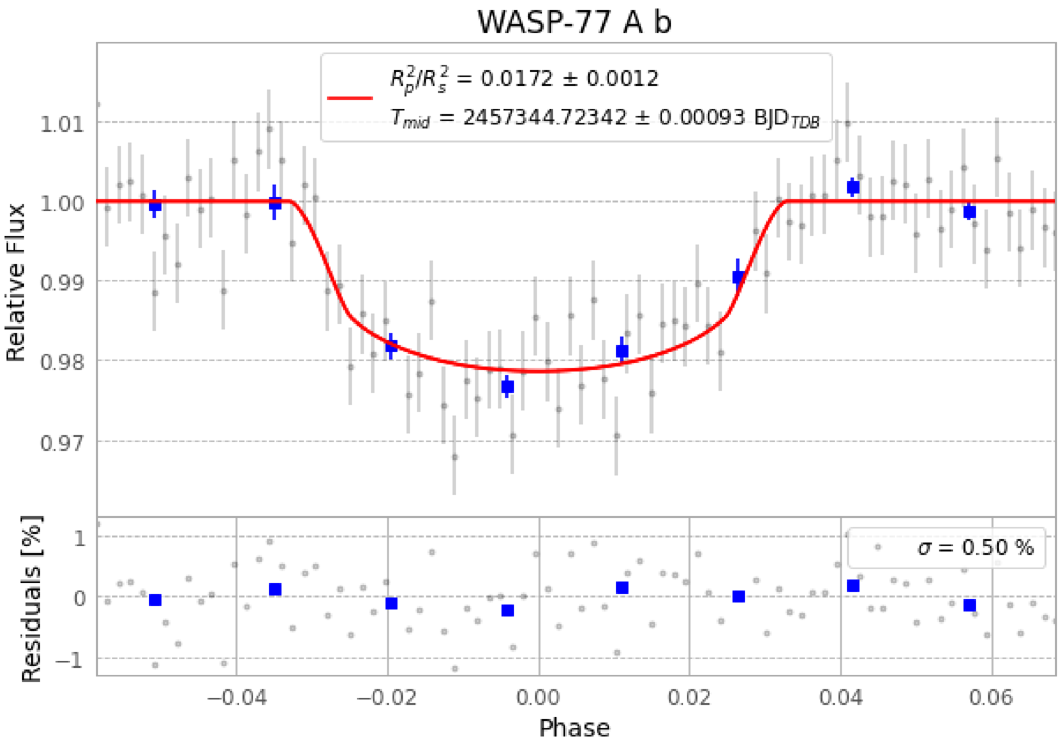

Figure 1 — FinalLightCurve_WASP-77 A b_2015-11-17.png (SHA-256 1fd438d663a6d39e…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [160, 222], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 87bc267937a9599b…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

84 FITS frames (55 MB), `WASP-77151118031812.FITS` → `WASP-77151118072712.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-77-151118.log (738f917d7390…), FinalLightCurve_WASP-77 A b_2015-11-17.png (1fd438d663a6…), FinalLightCurve_WASP-77 A b_2015-11-17.csv (5785d098357d…)

Compute resources

Wall time 140.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)